STAFF REVIEW · H446 071 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Estimate guiding choices, not necessarily guaranteeing optimum.

2. When the branch cannot produce a valid result.

3. Completed items per unit time.

4. Q becomes 0; D is sampled at the active clock edge.

I Do — source worked steps

1. Explain — Start a new task each cycle while earlier tasks advance; last task finishes cycle4.

We Do — guided CHECK

1. Explain
This is a throughput model assuming one task enters each cycle and no stall.

You Do — source mark schemes

1. T1 occupies stages at cycles1,2 [2]; T2 at cycles2,3 [1]; T3 at cycles3,4 [1]; finishes cycle4 [2].

2. Pipeline $4+6-1=9$ cycles [3]; sequential $4\times6=24$ [2]; assumes no stall [1].

3. Backtracking maze [2], data mining shopping [2], visualisation charts [2], pipelining overlapping stages [2].

Exit Ticket — source answer

1. False. Overlap typically improves throughput.